

# Lecture 1:

## What is quantum computing, and why should we care?

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Universidade do Minho



**Quantum Data Science**  
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# Plan

- Grave* Why does quantum matter?
- Adagio* Computing with quantum mechanics
- Andante* Quantum programs and quantum computers
- Allegro* Quantum information
- Vivace* What's next?

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# *Grave*

Why does quantum matter?

# Why quantum computing?

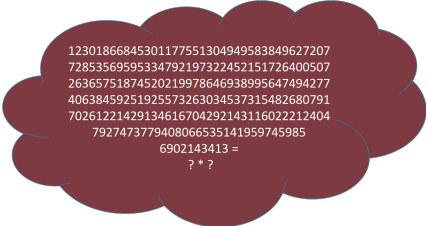
## Why should we care?

In simple terms, because a lot of really difficult, complex problems will forever remain **out of reach** of classical supercomputers ...

- Classical computer technology is running up against **fundamental size limitations**
- Quantum computing brings new **physical resources** to handle some really hard problems.

# Why quantum computing?

## Prime factorization



12301866845301177551304949583849627207  
72853569595334792197322452151726400507  
26365751874520219978646938995647494277  
40638459251925573263034537315482680791  
70261221429134616704292143116022212404  
7927473779408066535141959745985  
6902143413 =  
? \* ?

- As factoring a 130 digit number takes around one month in a massively parallel computer network, a 400 digit number will take about the age of the universe ( $10^{10}$  years).
- However, Shor's algorithm achieves this in polynomial time.

# Moving fast ...

## Quantum is moving (very) fast...

Research on quantum technologies is **speeding up**, and has already **created the first operational and commercially available applications**.

## The quantum decade: the 2020s ...

- For the first time the viability of quantum computing is **demonstrated in a number of problems** and **its utility discussed across industries**.
- Efforts, at national or international levels, to further **scale up** this research and development are in place.
- A **cross-industry race has begun** to secure quantum talent, build quantum skills, map real-world problems to quantum algorithms, and capture quantum application intellectual property.

... but still uncertain what the future will bring

- Quantum computing will have a **substantial impact on societies**:
- ... indeed, questions on its impact have changed from **if** to **when** and **how**,
- ... even if its **commercial potential** in the near term is still debatable.

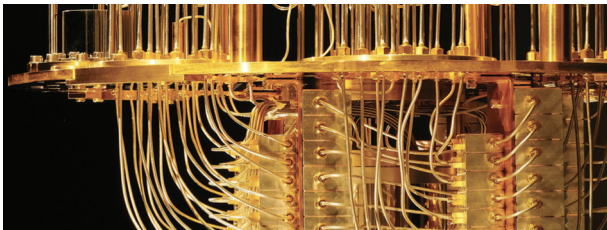
What we know for sure

- Resorting to a so **radically different technology**, it is difficult to **anticipate its evolution**.
- Emerging as an **entirely new paradigm**, quantum hardware and software **brings no similarity to their classical counterparts**.



## ... but what are we talking about?

- **Information** has a crucial **physical** dimension
- **Computation** is always a **physical** process



# Quantum Mechanics meets' Computer Science

Two main intellectual achievements of the 20th century met

- Computer Science and Information theory progressed by **abstracting** from the physical reality. This was the key of its success to an extent that **its origin was almost forgotten**.
- On the other hand, **quantum mechanics** ubiquitously underlies ICT devices at the implementation level, but had no influence on the **computational model** itself ...
- ... until **now!**

# Quantum Mechanics 'meets' Computer Science

Alan Turing (1912 - 1934)



*On Computable Numbers, with an Application to the Entscheidungsproblem* (1936)  
(computability and the birth of Computer Science)

# Quantum Mechanics 'meets' Computer Science

Richard Feynman (1918 - 1988)



*Simulating Physics with Computers* (1982)  
(quantum reality as a computational resource)

# Quantum effects as computational resources

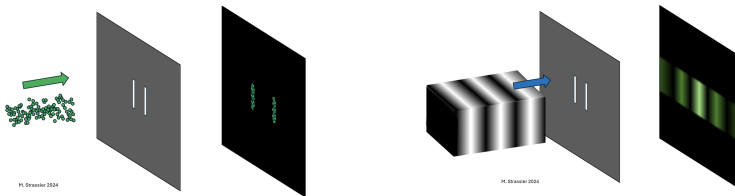
## Superposition

Our perception is that an object — e.g. a **bit** — exists in a well-defined state, even when we are not looking at it.

**However:** A quantum state **holds information of both possible classical states**, collapsing to one of them upon **observation**.



# Superposition: The double slit experiment



The wave-like pattern appears even when a **single photon** is emitted: it somehow **takes both paths** and **interacts with itself**.

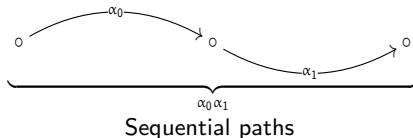
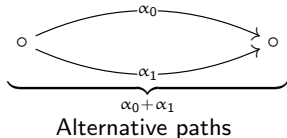
**What can be said about this weird behaviour?**

Just this: the rules governing the location where the photon will be found have the structure of wave interference.

# What does it mean?

Let's call upon Physics, as computation is a physical process:

- **Physics** identify the system's structure and assign numerical **probabilities** to each possible path,
- and classically the language of **probability theory** captures **its physical evolution**.

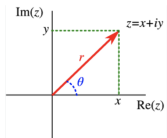


(Kolmogorov, 1903-1987, additivity axioms)

## What does it mean?

Quantum phenomena, however, cannot be described this way:

Paths are labeled by **complex** numbers ( $\alpha = a + bi$  or  $re^{i\theta}$ ), called their **amplitudes**. Thus, they are not scalars, but have a **direction**.



So, when added, they interfere, **canceling** or **reinforcing** each other.

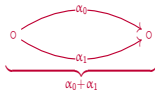
Probabilities are recovered as the **norms squared** of amplitudes:

$$\text{Born's rule } p = |\alpha|^2 = \bar{\alpha}\alpha$$

(for Max Born, 1882-1970)

# Superposition

Let's compute the total probability in



$$\begin{aligned} p &= |\alpha_0 + \alpha_1|^2 \\ &= \overline{(\alpha_0 + \alpha_1)}(\alpha_0 + \alpha_1) \\ &= (\overline{\alpha_0} + \overline{\alpha_1})(\alpha_0 + \alpha_1) \\ &= |\alpha_0|^2 + |\alpha_1|^2 + \overline{\alpha_0}\alpha_1 + \alpha_0\overline{\alpha_1} \\ &= p_0 + p_1 + |\alpha_0||\alpha_1| \left( e^{i(\theta_1 - \theta_0)} + e^{-i(\theta_1 - \theta_0)} \right) \\ &= p_0 + p_1 + \underbrace{2\sqrt{p_0 p_1} \cos(\theta_1 - \theta_0)}_{\text{interference}} \end{aligned}$$

(expressing  $\alpha_j$  in polar form  $|\alpha_j|e^{i\theta_j}$  and resorting to  $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ )

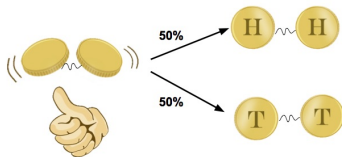
Depending on  $\theta_1 - \theta_0$ , **interference** can be either **negative** or **positive**. If the system's evolution depends only on that difference, then the system **must have**, somehow, **experienced both paths**.

# Quantum effects as computational resources

## Entanglement

Our perception is that objects are directly affected only by nearby objects, i.e. the laws of physics work in a local way.

**However:** two qubits can be so strongly correlated, or **entangled**, that an action performed on one of them **can have an immediate effect on the other** even at a distance.



Quantum entanglement can be used to create instantaneous agreement on information over distance (*teleportation*).

# Warning: Entanglement destroys interference

## Add detectors to each slit

The two systems (photon and detector) become entangled:

(photon | **left** obs)      (photon | **right** obs)

and the **classically expected path** re-appears:



Through the detectors, the universe **records which path was followed** and the availability of **this information determines the physical outcome**.

Mathematically, ...

- **No path information recorded**: sum amplitudes (entailing interference) and then compute the probability (quantum behaviour)
- **Path information recorded**: compute the probability of each path and sum them (classical behaviour)

## Warning: Entanglement destroys interference

This is **why**

- Classical behaviour seems **the norm in our daily experience**.

The classical world is just quantum reality  
**seen through the blurring effect of trillions of interactions**  
with the environment.

- Quantum states are critically **unstable** ... and have to be kept in **isolation**
- ... and quantum computers are so **hard** (and so **costly**) to build.

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## *Andante*

Quantum programs and quantum computers

# Quantum computational resources

## Superposition

which allows a quantum memory to hold information of many classical states "simultaneously", each of them carrying a specific weight, or amplitude, which dictates the probability of becoming manifest upon observation. This entails a form of **natural parallelism**.

## Interference

... through which different superposed states combine in a wave-like fashion with positive and negative amplitudes **canceling** or **reinforcing** each other.

## Entanglement

... witnessing **strong correlations** between different parts of a system such that an action performed on one of them can have an **immediate** effect on the other, even **at an arbitrary long distance**.

# Can we play the quantum game with a classical computer?

Do to superposition, a single quantum state (qubit) has a **continuum of possible values**, potentially storing lots of classical data.

Simulating a computation with qubits in a classical computer would be extremely hard, i.e. extremely inefficient as the number of qubits increases:

- For 100 qubits the state space would require to store  $2^{100} \approx 10^{30}$  complex numbers!
- And what about rotating a vector in a vector space of dimension  $10^{30}$ ?

Thus, the way ahead is

to resort to **quantum reality as a computational resource**

Richard Feynman, *Simulating Physics with Computers* (1982)

... but quantum computing requires engineering!

If superposition entails some form of **natural parallelism**, making possible the simultaneous computation of all solutions to a problem (all paths), **such a magic is refrained** by a basic fact: when reading the outcome of such a process,

only one solution is picked up at random, all the other discarded.

No magic, but engineering

Quantum computation = quantum resources + engineering

quantum programming amounts to **engineering** this potential **parallelism** with suitable forms of **interference** to come up to the **relevant outcomes** with **high probability**.

# Quantum Computation

David Deutsch (1953)



The Church-Turing principle and the universal quantum computer (1985)  
(first example of a quantum algorithm that is exponentially faster than any possible deterministic classical one)

# Quantum Computation

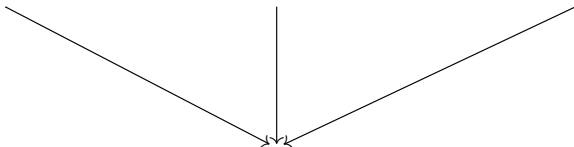
*quantum resources*



*quantum algorithms*



*computability*



# What can be expected from quantum computation?

- The meaning of **computable** remains the same ...
- ... but the order of **complexity** may change

## The landmark

Factoring in **polynomial** time -  $\mathcal{O}((\ln n)^3)$

Peter Shor, *Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer* (1994)

## ... major steps ...

- 1994: Peter Shor's factorization algorithm (exponential speed-up),
- 1996: Grover's unstructured search (quadratic speed-up),
- 2018: Advances in hash collision search, i.e finding two items identical in a long list — serious threat to the basic building blocks of secure e-commerce.
- 2019: Google announced to have achieved quantum supremacy. The problem considered two random number generators, each producing a sequence of digits, trying to determine whether both sequences are completely independent from each other, or are related in a hidden way.
- ... followed by similar announcements in China.
- Explosion of emerging applications in several domains: security, finance, optimization, machine learning, etc., and huge, global research investments in quantum hardware, communications, sensors, and algorithmics.

## Huge potential, but moderate expectations

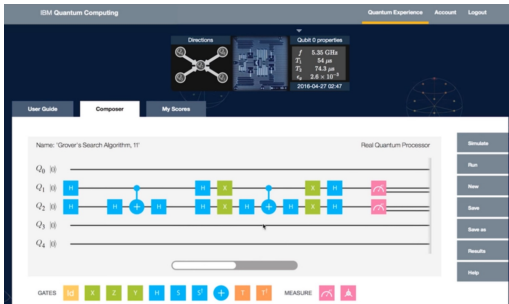
- Quantum systems are **highly susceptible to environmental disruptions** ... entailing **physical challenges** in the design of quantum computers that are both expensive and difficult to address.
- Coexistence of several **alternative architectural models and physical implementation of qubits**; no clear winner.
- Quantum **software**, as a true engineering discipline, is **still in its infancy**: new methods, new mathematics, new algorithmic techniques are in order.
- As of today, most of the applications remain theoretical and/or not scaled for market rollout.
- Quantum hardware and software production is **extremely costly** and restricted to a relatively small number of companies (although within a vibrant ecosystem)

Availability of **proof of concept** hardware

# Where exactly do we stand?

## Typical configurations & business models

- the quantum device as a coprocessor
- typically accessed as a service over the cloud



# Where exactly do we stand?

## NISQ - Noisy Intermediate-Scale Quantum Hybrid (J. Preskill, 2028)

- Machines **large enough** to be interesting (roughly 50 to 1 000 qubits) but **too error-prone**.
- Each operation introduces **accuracy errors** in the system, which **limits the size of quantum circuits** that can be executed reliably.
- NISQ machines run real applications, but their practical advantage over classical computers **remains limited or unproven**.

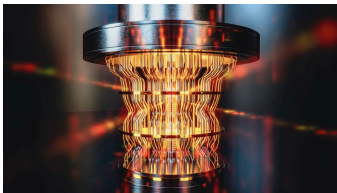
### Can run:

- Variational algorithms with short circuits in classical-quantum loops.
- Quantum chemistry approximations for small molecules.
- Optimization heuristics and machine learning experiments with quantum kernel methods.

## Where exactly do we stand?

Fault-tolerant machines (currently emerging!)

- Key technology: **quantum error correction**, which encodes quantum information across **multiple physical qubits** to create **logical qubits** that can detect and correct errors in real time  
(1 logical qubit may require 1000 physical ones)
- Can run **arbitrarily long computations** to completion regardless of how many gates are required, because errors are continuously corrected faster than they accumulate
- Exciting applications, but requiring **from hundreds to millions of logical qubits**, depending on the problem and the expected precision



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# Quantum Information

# The *motto*

Information is a physical entity. Any computation is a physical process

Thus, any abstract computational model has always to reflect **what we know about reality**, and **quantum theory** is our **current best tool** in this road.

## Three ways to represent information

- ... in a **two-state** device
- ... in a similar way but with **uncertainty**
- ... in a **quantum** device

## Binary information

Computational / information states based on Boolean values 0 and 1 which can be represented by **vectors**:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

If rows are labelled from 0 onwards, the presence of 1 in a cell identifies the number represented by the vector.

Larger state spaces are built with the (Kronecker) **tensor** product:

$$\begin{bmatrix} p_0 \\ p_1 \end{bmatrix} \otimes \begin{bmatrix} q_0 \\ q_1 \end{bmatrix} = \begin{bmatrix} p_0 \begin{bmatrix} q_0 \\ q_1 \end{bmatrix} \\ p_1 \begin{bmatrix} q_0 \\ q_1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} p_0 q_0 \\ p_0 q_1 \\ p_1 q_0 \\ p_1 q_1 \end{bmatrix}$$

# Binary information

## Examples

$$|00\rangle = |0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|4\rangle = |100\rangle = |1\rangle \otimes |0\rangle \otimes |0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

# Binary information

## Operations as matrices

$$I(x) = x$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$X(x) = \neg x$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\underline{1}(x) = 1$$

$$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\underline{0}(x) = 0$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} I|0\rangle &= |0\rangle & I|1\rangle &= |1\rangle \\ X|0\rangle &= |1\rangle & X|1\rangle &= |0\rangle \\ \underline{1}|0\rangle &= |1\rangle & \underline{1}|1\rangle &= |1\rangle \\ \underline{0}|0\rangle &= |0\rangle & \underline{0}|1\rangle &= |0\rangle \end{aligned}$$

# Binary information

## Composition

Sequential composition: **matrix multiplication**

Parallel composition: **Kronecker product**  $\otimes$

$$M \otimes N = \begin{bmatrix} M_{1,1}N & \cdots & M_{1,n}N \\ \vdots & & \vdots \\ M_{m,1}N & \cdots & M_{m,n}N \end{bmatrix}$$

for example

$$X \otimes \underline{1} \otimes I |101\rangle = X \otimes \underline{1} \otimes I (|1\rangle \otimes |0\rangle \otimes |1\rangle) = X|1\rangle \otimes \underline{1}|0\rangle \otimes I|1\rangle = |011\rangle$$

## Probabilistic information

... the system is **always in some well defined state**, even if we do not know which:

**State**: is a **vector of probabilities** in  $\mathcal{R}^n$

$$[p_0 \cdots p_n]^T \text{ such that } \sum_i p_i = 1$$

which express **indeterminacy** about the system's exact physical configuration

**Operator**: is a **double stochastic** matrix where  $M_{i,j}$  specifies the probability of evolution from state  $j$  to  $i$

# Quantum information is a different story

A quantum state holds the information of **both** possible classical states:



A unit of information lives in a 2-dimensional **complex** vector space:

$$|\nu\rangle = \alpha|0\rangle + \beta|1\rangle$$

and thus possesses a **continuum of possible values**, so potentially, can store lots of classical data.

# Quantum information is a different story

However, all this potential is **hidden**:

when **observed**  $|v\rangle$  **collapses into a classic state**:  $|0\rangle$ , with probability  $|\alpha|^2$ ,  
or  $|1\rangle$ , with probability  $|\beta|^2$ .

(Recall:  $|\alpha| = \sqrt{\alpha\bar{\alpha}}$  for a complex  $\alpha$ )

The outcome of an observation is **probabilistic**, which calls for a restriction to **unit** vectors, i.e. st

$$|\alpha|^2 + |\beta|^2 = 1$$

to represent quantum states.

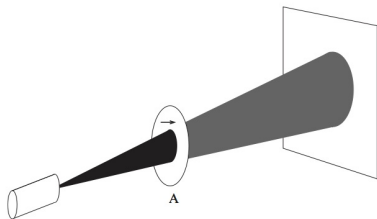
# Quantum information is a different story

This quantum state is **not** a probabilistic mixture: it is **not** true that the state is really either  $|0\rangle$  or  $|1\rangle$  and we just do not happen to know which.

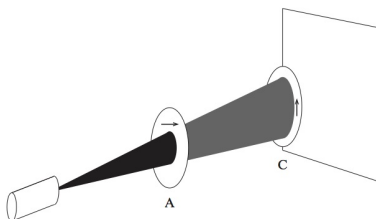
**Amplitudes are not real numbers** (e.g. probabilities) that can only increase when added, but **complex** so that they can **cancel each other or lower their probability**, thus capturing a fundamental **quantum resource**:

**interference**

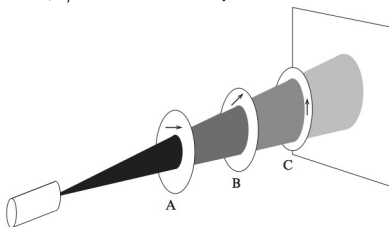
# Quantum information: An experiment with a photon



$|0\rangle$  - horizontal polarization



$|1\rangle$  - vertical polarization



(from [Reifell & Polak, 2011])

# Quantum information: An experiment with a photon

## An explanation for a *single* photon experiment

- The photon's polarization state is modelled by a unit vector, for example the following **linear combination of  $|0\rangle$  and  $|1\rangle$** :

$$\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

which corresponds to a polarization of 45 degrees.

- On passing a polaroid the photon will be absorbed or leave the polaroid with its polarization aligned with the polaroid's axis.
- The probability to go through the polaroid is the **square of the magnitude** of the amplitude of its polarization in the direction of the polaroid's axis.

## Quantum information: An experiment with a photon

- Polarization of polaroid B is  $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$
- This vector, together with  $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$  forms a **basis** for the 2-dimensional vector space
- of which  $\{|0\rangle, |1\rangle\}$  is another one (the so-called **computational basis**).
- Expressing  $|0\rangle$  in terms of  $|+\rangle$  and  $|-\rangle$  yields

$$|0\rangle = \frac{1}{\sqrt{2}}|-\rangle + \frac{1}{\sqrt{2}}|+\rangle$$

which explains why a visible effect appears in the wall:

the photon goes through C with 50% of probability (i.e.  $|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$ ).

# A unit of quantum information

Photon's polarization **states** are represented as unit vectors in a **2-dimensional complex vector space**, typically as a

non trivial linear combination  $\equiv$  **superposition** of vectors in a basis

$$|v\rangle = \alpha|0\rangle + \beta|1\rangle$$

A basis provides an **observation** (or **measurement**) tool, e.g.

$$\bigcirc \frown \bigcirc = \{|0\rangle, |1\rangle\} \quad \text{or} \quad \bigcirc \frown \bigcirc = \{|-\rangle, |+\rangle\}$$

# A unit of quantum information

Observation of a state

$$|v\rangle = \alpha|u\rangle + \beta|u'\rangle$$

transforms the state into one of the basis vectors in

$$\bigcirc \frown \bigcirc = \{|u\rangle, |u'\rangle\}$$

with a probability given by the norm square of its amplitude.

In other (the quantum mechanics) words:

measurement collapses  $|v\rangle$  into a classic state

A subsequent measurement wrt the same basis returns  $|u\rangle$  with probability 1.

# Qubits

The space of possible polarization states of a photon is an example of a

quantum bit (qubit)

as does any quantum system (e.g. a electron spin or an atom) that can be modelled by a two-dimensional complex vector space.

- In practice it is not yet clear which two-state systems will be most suitable for physical realizations of qubits: it is likely that a variety of physical representation will be used.
- and they are fragile and unstable which entails the need for qubits' strong isolation, typically very hard to achieve.

# Qubits

A qubit has ... a **continuum of possible values**

- potentially, it can store lots of classical data
- but the amount of information that can be extracted from a qubit by measurement is severely **restricted**: a single measurement yields at most a single classical bit of information;
- as measurement changes the state, **one cannot make two measurements on the original state** of a qubit.
- as an unknown quantum state **cannot be cloned**, it is not possible to measure a qubit's state in two ways, even indirectly by copying its state and measuring the copy.

*Vivace*

## What's next? ...

# This Curricular Unit

`http://lmf.di.uminho.pt/qu4DataSci-2627`